

Heeseung Choi

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Research Interests

My research interests include, but are not limited to:

- RF/Microwave Circuits & Front-End Systems
- Applied Electromagnetics
- Wireless Communication

Education

Yonsei University, Seoul, South Korea

03/2020 - 02/2026

Bachelor of Engineering in Electrical and Electronic Engineering

GPA: 4.07/4.30

University of Georgia, GA, United States

08/2024 - 12/2024

Exchange Program

GPA: 4.00/4.00

Research Experience

Undergraduate Researcher

Seoul, South Korea

Yonsei University — Advisor: *Prof. Byung-Wook Min*

01/2025 – Present

- Led an undergraduate capstone on a 1-bit Phase Shifting Switch Module for Reconfigurable Intelligent Surface, optimizing transistor sizing and matching networks to achieve low insertion loss and a stable 180° phase shift over a wide bandwidth.
- Awarded the IEEE AP-S Undergraduate Research Scholarship; presented at KIEES (2025).
- Currently extending the 1-bit design to 1.5-/2-bit switch modules for higher phase resolution and improved beam control; optimizing phase accuracy, insertion loss, and an 8-channel layout for dual-polarized RIS integration, with prototypes entering fabrication.

Publications & Presentations

- [1] **Heeseung Choi** and Byung-Wook Min, “1-bit Phase-Shifting Switch Module for Reconfigurable Intelligent Surfaces,” *KIEES Summer Conference*, Pyeongchang, South Korea, Aug. 2025 (Poster Presentation).

Awards & Scholarships

- **IEEE AP-S Undergraduate Summer Research Scholarship**
IEEE Antennas and Propagation Society
- **Korea-U.S. Technology Fields Exchange Program Scholarship**
Korea Institute for Advancement of Technology
- **Yonsei University Semiconductor Track Program**
Yonsei University
- **High Honors (Dean's List)** Spring 2020, Spring & Fall 2023, Spring 2024, Spring 2025
Yonsei University

Skills

- **Tools:** Keysight ADS, HFSS, MATLAB, PSPICE/LTSPICE, VIVADO
- **Relevant Coursework:**
 - **Electromagnetism Track:** Electricity & Magnetism I/II, Microwave Engineering, Microwave Experiment, Antenna Engineering
 - **Communication Track:** Communication Theory, Communication Systems, Digital Signal Processing, Signals & Systems
 - **Microelectronics Track:** Electronic Circuits I/II, Physical Electronics